SSH

What is SSH?

It's a safe way to make your local provide its identity without entering password every time

Components:

1_Public Key: can be shared

2-private key: cannot be shared

How it works:

When git push:

1-**GitHub sends an encrypted challenge** to your computer.
(It's basically saying: "Prove that you are who you claim to be.")

2- **Your computer uses your private key** to sign (encrypt) that challenge.

3- **GitHub uses your public key** (which you previously uploaded to your GitHub account) to verify the signature.

Commands

```
A-ONE@DESKTOP-IIOBSG2 MINGW64 ~
$ ssh-keygen -t ed25519 -C "ya43712@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/A-ONE/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/A-ONE/.ssh/id_ed25519
Your public key has been saved in /c/Users/A-ONE/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:X6nY29aEdUXYK1fCAtXdFvzAV9fkoUAqzCfRB32dRnA ya43712@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|       ...==.==E@|
|      o ...oo.X*%|
|       = o. .o.=*|
|        +    o.oo|
|        S   ooo. |
|         + o. .  |
|        . +  o   |
|           o. .  |
|           ...   |
+----[SHA256]-----+

A-ONE@DESKTOP-IIOBSG2 MINGW64 ~
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICHutbMdfHvWMKB5fLaMUkBwT4Re1KBO25h1RKEsQfWo
 ya43712@gmail.com

A-ONE@DESKTOP-IIOBSG2 MINGW64 ~
$ ssh -T git@github.com
Hi Yasser213-cpu! You've successfully authenticated, but GitHub does not provide
 shell access.

A-ONE@DESKTOP-IIOBSG2 MINGW64 ~
$ |
```